

A Guide to the Summer 2025 Cognitive Study

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1 Overview

The impact of temperature on cognitive performance has significant implications for a wide range of occupations susceptible to heat stress, such as firefighters, athletes, soldiers, healthcare workers, and pilots. Prior research suggests that an increase of roughly 1 °C in core temperature is enough to impact performance on a variety of cognitive tasks.

My hypothesis was that when core body temperature is elevated, individuals will perform worse on a cognitive battery featuring three distinct tests: the Balloon Analogue Risk Test (BART), the Psychomotor Vigilance Test (PVT), and the Trail Making Test (TMT).

2 Initial Stress

2.1 Design and History

This study used a within-subjects design (Figure 1, 2) with 11 college-aged (9 male, 2 female) adults who participated in both an experimental and control trial. In the experimental trial, participants dressed in insulative personal protective suits walked on a treadmill in a 40 °C environment until their core temperature reached 38 °C. Meanwhile, in the control trial, participants walked for a similar duration of time, but without PPE.

The order of trials was randomized, so that roughly half of the participants did the control trial first and the other half did the experimental trial first. If the participant was assigned to do the experimental trial first, I would take note of the duration it took them to reach 38 °C. They would then walk the same duration in the control trial.

For the participants who did the control trial first, I ran into a dilemma. Since the participant had not done the experimental trial yet, it was unknown how long it would take them to reach 38 °C. When I was first testing this current protocol, I found that on average, most participants took 20 minutes to reach 38 °C. Thus, for those who did the control trial first, I had them walk on the treadmill for 20 minutes. Although the duration appeared to be slightly different, this difference was not statistically significant (Table 1).

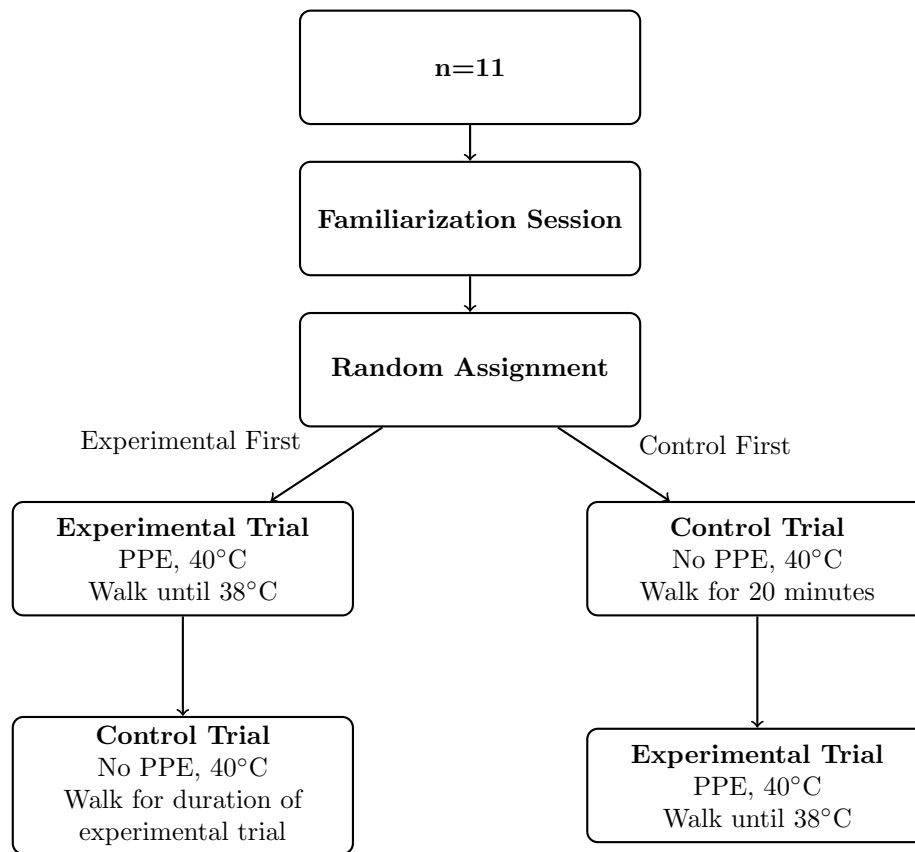


Figure 1: Participants completed a familiarization session before being randomly assigned to begin with either the experimental or control trial, crossing over to the alternate condition afterward.

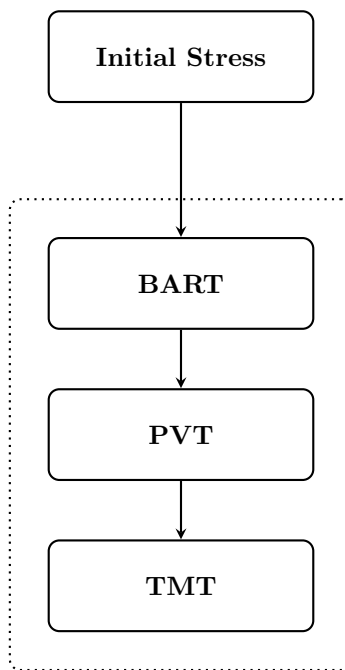


Figure 2: After the initial stress, participants completed the cognitive battery in the following order: BART $\rightarrow$ PVT $\rightarrow$ TMT.

Condition	Duration (min)	Tympanic Temperature (°C)
Control	20.6 $\pm$ 5.04	36.9 $\pm$ 0.337
PPE	24.2 $\pm$ 4.07	37.8 $\pm$ 0.279

Table 1: Duration of the initial stress was not significantly different between the two groups. Tympanic temperature differed significantly between the two groups by ~ 1 °C.

I actually explored many different protocols before settling on this current protocol. I found early on that merely sitting in the room with PPE on was impractical for this type of experiment. One of our lab members, Olek, spent over one hour and thirty minutes in the hot room. In addition to the duration of the cognitive battery, his trial took nearly 2 hours in the hot room. Not only would it be impossible to recruit volunteers if each trial was an hour and a half in the hot room, I also concluded that spending such a long time in the hot room would introduce additional confounding variables: namely, the boredom of spending so long in the hot room, as well as the anxiety of watching yourself slowly heat up. These confounding variables would make it very difficult to isolate solely the impact of core body temperature on cognition.

Another protocol that was proposed was using a bath tub and heating people using water. Relative to air, water has a higher specific heat capacity. In other words, at 40 °C, a liter of water holds a lot more thermal energy than a liter of air. Practically, this means water has a lot of thermal energy to transfer to you. Thus, you heat up a lot faster in 40 °C than in 40 °C air. This is actually why scientists use water baths to thaw media and cells. While this was certainly a good idea to avoid the problems with the prior protocol, it was also more logistically difficult, and I opted to find a simpler solution.

Finally, I tested out the current iteration of the protocol. In my early tests, I found that light walking on the treadmill significantly accelerated a participant’s increase in core body temperature without causing significant fatigue. For most people, it took only 20 minutes to get to a tympanic temperature of 38 °C.

2.2 Metrics and Data Analysis

The results from measuring tympanic temperature showed that the initial stress in the experimental trial significantly raised temperatures by 1 °C relative to the control trial (Table 1 and Figure 3). More data can be found in Supplementary Table 1.

3 Balloon Analogue Risk Test

3.1 Design and History

The Balloon Analogue Risk Test (BART) is a test designed to measure risk-taking behavior. It was originally developed around 25 years ago, and has since

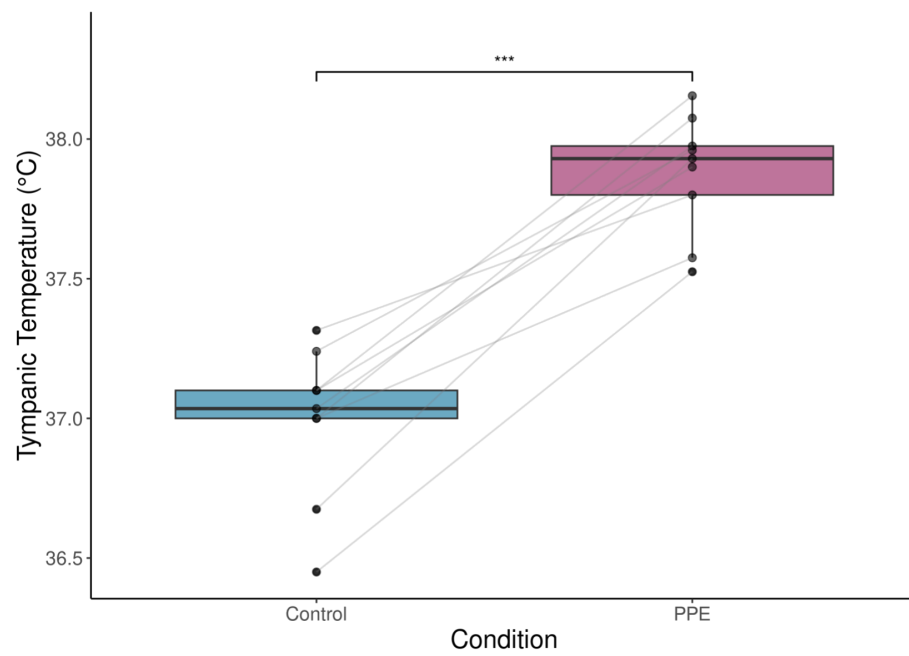


Figure 3: Highly significant increase in tympanic temperature, a common and reliable index for core temperature.

gone through a variety of iterations. When designing this test, I considered many of these different iterations and eventually settled on what is known as the automatic BART, primarily because of this excellent paper (Burst Beliefs). I also added an additional choice to the test, which notably had never been done before, which I called the "Top Off."

The basic gist of the test is that across 30 trials (i.e. 30 different balloons), the participant is tasked with pumping up the balloon as much as possible without popping it (Figure 4). Each balloon has a precomputed, hidden, and pseudo-random number of pumps at which it will pop (pop point). The participant chooses a pump amount with a slider, then presses PUMP once; pumps are then simulated automatically. Once a participant clicks PUMP, the participant cannot alter their choice. This is why we call it the "automatic" BART.

Each simulated pump adds \$0.01 to a temporary bank. If the balloon pops before collection, the user earns \$0.00 for that trial. If the balloon does not pop by the end of the trial, the money from the temporary bank is transferred to total earnings.

Behind the scenes, this test has a lot of additional considerations. The precomputed number for when a balloon will pop is not completely random. The rationale is that while each individual trial should be as random as possible, the *average* pop point across all trials should be roughly equal every time the test is given. For example, let's say we implemented the test and each trial, the number was randomly picked between 1 and 140. Because each trial is technically independent of one another, it is entirely possible for a randomly generated test to yield a sequence of unusually early pops or unusually late pops purely by chance. In other words, one participant might face balloons that pop at an average of 40 pumps, while another faces balloons that pop at an average of 100 pumps. This would make it impossible to actually compare these two sessions, because these are fundamentally two different circumstances.

To solve this, I designed the test so that trials are chunked into blocks of 10 (for 30 trials $\rightarrow$ 3 blocks). Within each block, the pop point of the balloon was sampled from a normal distribution centered around a target average. Then, the program nudges values so the sum of the pop point across all trials in a block equals the target average * block size.

As I mentioned previously, I added a novel "top off" mechanism to my version of the BART (Figure 5). The rationale behind this is that the automatic BART, although being the best iteration of the BART, loses some level of granularity because the participant is making one choice per trial, as opposed to 60+ choices per trial. Before the test starts, each trial is marked as Top Off eligible or not. By default, 15/30 are eligible. If the participant completes the initial pump session without popping the balloon and if that trial is Top Off eligible, the participant is given the option to add 0–9 extra pumps to the balloon. It is important to note that the pop point for the balloon does not change, so if adding extra pumps to the balloon exceeds the pop point, the participant will receive \$0.00 for that trial.

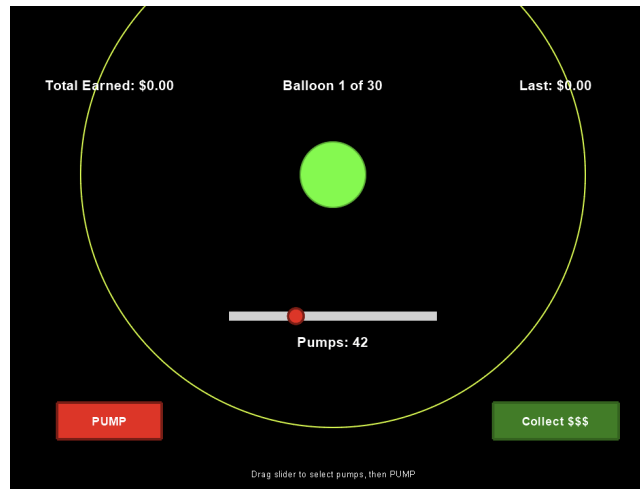


Figure 4: An example of a trial from the BART

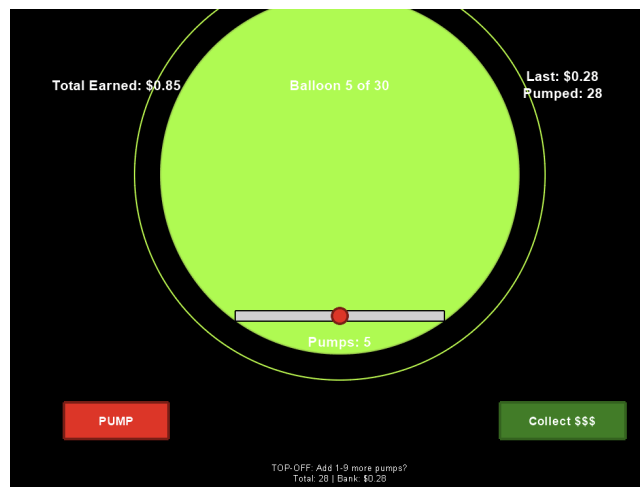


Figure 5: The Top Off mechanism from the BART

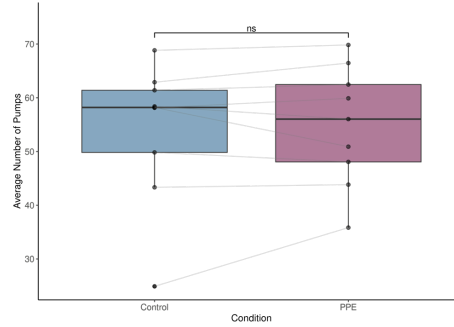


Figure 6: No significant difference with respect to average number of pumps, the primary measure of the PVT, indicating no impact on risk-taking behavior.

3.2 Metrics and Results

Average number of pumps is the primary measure of the BART, but a variety of other metrics can be found in Supplementary Table 3 and 4. When I ran the statistics, I found no significant difference between conditions with respect to average number of pumps (Figure 6) or any of the other secondary measures. One of the reasons this test may not be extremely valid is because there is no reward linked with performance. In future studies, I would highly recommend finding a way to link some form of a reward to performance.

4 Psychomotor Vigilance Test

4.1 Design and History

The Psychomotor Vigilance Test (PVT) is a relatively simple test used to measure vigilance and alertness. On each trial, participants viewed a blank black screen during a variable inter-stimulus interval (ISI), after which a red millisecond counter appeared in the center of the screen (Figure 8). Participants were instructed to press the spacebar as quickly as possible upon the onset of this counter.

Responses made during the ISI were classified as false starts (Figure 9). If no response occurred before the timeout (3000 ms), the trial was recorded as no response. ISIs were sampled from a uniform distribution (2-10s). In total, the test lasted 5 minutes.

The PVT also counts the number of “lapses,” which I defined relatively arbitrarily as response times (RTs) exceeding 500ms, as well as “no-response timeouts,” which is when the user fails to respond after 3000 ms.

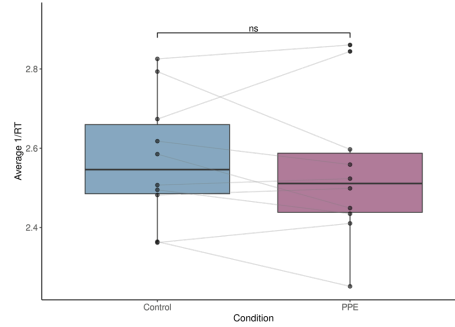


Figure 7: No significant difference with respect to the reciprocal transform of reaction time, the primary measure of the PVT, indicating no impact on alertness.

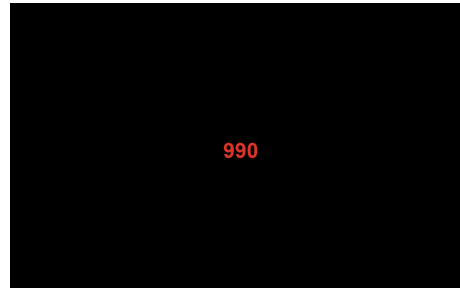


Figure 8: An example of a trial from the PVT

4.2 Metrics and Results

The reciprocal transform of reaction time is the primary measure of the PVT, and prior literature has found this to be more reliable than reaction time without the reciprocal transform. When I ran the statistics on this metric (Figure 7) as well as other secondary measures (listed in Supplementary Table 5 and 6), there was no statistically significant difference.

5 Trailmaking Test

5.1 Design and History

The Trailmaking Test (TMT) is a classic cognitive test used in a variety of neuropsychological testing settings that measures two key cognitive functions: set-shifting (and implicitly, working memory) and visual search.

On each trial, stimuli were displayed as circles containing numbers, letters or geometric shapes, and participants advanced by clicking items in the required sequence. The stimuli were randomly distributed across the page (Figure 10).



Figure 9: An example of the false start message from the PVT

The introduction of a third category, geometric shapes, is one of the key modifications I have made to the TMT commonly described in the lecture. The order of geometric shapes is determined by the number of sides (triangle, square, pentagon, etc). This third category serves to make the test more difficult.

The sequence for a particular trial always involves the participant alternating between the three categories in a particular order (e.g. shapes $\rightarrow$ letters $\rightarrow$ numbers), and can either be ascending or descending. For example, if the participant is given shapes $\rightarrow$ letters $\rightarrow$ numbers in an ascending manner, the order of sequence is triangle $\rightarrow$ A $\rightarrow$ 1 $\rightarrow$ square $\rightarrow$ B $\rightarrow$ 2 $\rightarrow$ pentagon $\rightarrow$ C $\rightarrow$ 3, and so on. If given in a descending manner, it would be the opposite direction.

When I ran the test last summer, every single run of the experiment would feature this sequence, in a fixed order:

1. Ascending; numbers $\rightarrow$ shapes $\rightarrow$ letters
2. Descending; numbers $\rightarrow$ shapes $\rightarrow$ letters
3. Ascending; shapes $\rightarrow$ letters $\rightarrow$ numbers
4. Descending; shapes $\rightarrow$ letters $\rightarrow$ numbers
5. Ascending; letters $\rightarrow$ numbers $\rightarrow$ shapes
6. Descending; letters $\rightarrow$ numbers $\rightarrow$ shapes

While I have left this in as an option, I have decided that randomizing the trials further would make this more rigorous of a test. The new mode generates all unique permutations of the three categories, shuffles the permutations, and runs each permutation ascending and then descending. This yields 12 trials.

I have found that the sweet spot for the difficulty of this test is that each trial has 6 stimuli per category. The descending trials, in particular, are very difficult. A participant will always know for an ascending trial that the first stimuli are 1, triangle, and A. However, in a descending trial, knowing the first stimuli is not very instinctive (am I looking for 6 or 5?). To help address this, the stimuli (in order) are displayed before each participant before the trial begins

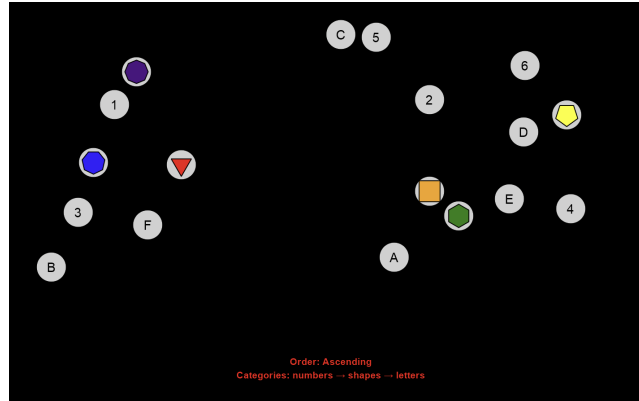


Figure 10: An example of a trial from the TMT

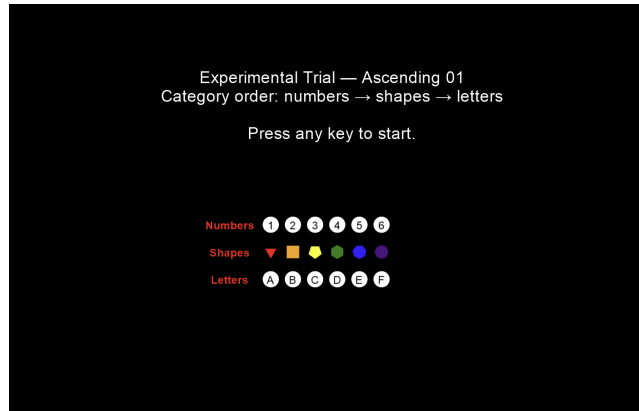


Figure 11: An example of the screen shown prior to a trial in the TMT

(Figure 11). It also helps to remind them the range of the stimuli ("Remember, the numbers range from 1 to 6, the letters range from A to F, and the shapes range from triangle to octagon"). While this test may seem easy, it is by far the most mentally taxing of the tests in the battery.

5.2 Metrics and Results

The results from the TMT were exciting. On the primary metric, average total duration (of a trial), there was a very significant increase in average total duration (Figure 12). There are two completely plausible reasons why someone might take longer on this test: firstly, it merely takes them longer to find the stimuli, or secondly, there are more wrong guesses (and each wrong guess contributed to a longer time). Statistical analysis on both these metrics found that they were both contributing factors (Figure 13 and 14). Note: in Figure

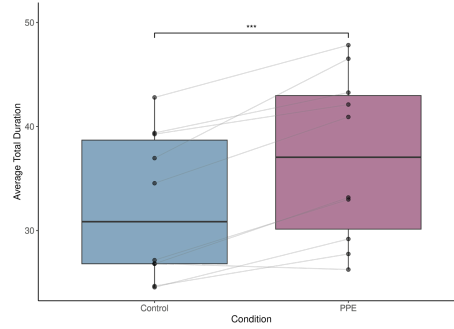


Figure 12: Significant increase in average total duration, the primary measure of the TMT.

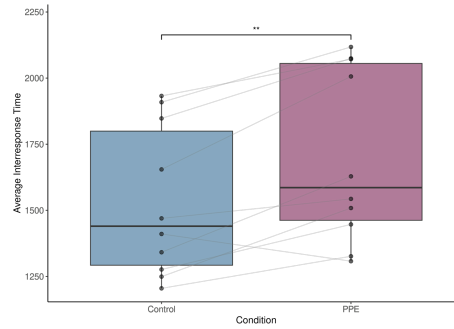


Figure 13: Average response time was significantly increased, which contributed to the result seen in Figure 12

13, in order to isolate the effect of *only* reaction time (and not include the effect of guessing wrong), any "connection" that involved a wrong guess before the correct response was removed. In other words, that figure is based on the participant's mean RT on correct, first-attempt responses only. Other secondary metrics are displayed in Supplementary Data Table 7.

6 Practical Information

6.1 How to Set up PsychTest

1. Install Python 3.11.9: <https://www.python.org/downloads/release/python-3119/>
2. During install, check "Add Python to PATH".
3. To check the install, open a terminal and type:

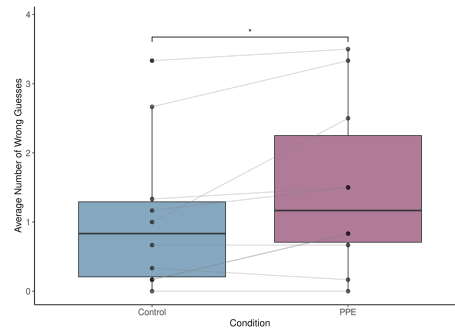


Figure 14: Average number of wrong guesses was significantly increased, which also contributed to the results seen in Figure 12

```
py -0p
```

You should see a 3.11 entry.

4. In your project folder:

```
cd C:\Users\-\Desktop\Psych
py -3.11 -m venv .venv
.venv\Scripts\activate
python -m pip install --upgrade pip setuptools wheel
python -m pip install -r requirements.txt
```

5. Run the app:

```
python master.py
```

6.2 How to Run PsychTest

1. Launch the battery:

```
python master.py
```

2. In the launcher, complete **Participant Info**:

- Enter **Participant ID** (required).
- Enter **Treatment / Condition**.

- Leave **Replay exact random sequence** OFF for normal data collection (enable only for reproducibility/debugging).
 - Optionally, enable self-report at beginning and/or end.
3. Configure **Screen Settings**:
 - Select full screen, and don't worry about changing resolutions.
 4. Configure **Test Order**:
 - Use the arrows to reorder tasks (BART, PVT, TMT).
 5. Verify **Task Parameters**. You may use default settings.
 6. Start the session:
 - Click **Run All Tests (in order)** for full battery, or **Run BART/Run PVT/Run TMT** for a single task.
 7. During task administration:
 - Press **ESC** to open the pause menu.
 - Pause menu options: **Resume**, **Quit Test**, or **Quit Battery**.
 - If paused, the entire test is frozen; the metrics are calculated assuming the pause never happened.
 8. Complete optional end self-report (if enabled), then confirm output:
 - A timestamped Excel workbook is written to:
 PsychSuite\Data\
 - Verify expected sheets are present (e.g., **BART**, **PVT**, **TMT**, **Derived Metrics**, **SessionLog**; plus **Self Report** if enabled).